

SeedHammer II — the taproot pathological wallet

The hardest wallet this constellation describes, round-tripped on the machine.

This is the four-tier degrading vault — the wallet `CORPUS.md` calls *the* pathological example — expressed as a **depth-3 taproot tree**:

```
tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,{and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,@0/48'/0'/0'/2'<0;1>/*,@1/48'/0'/1'/2'<0;1>/*,@2/48'/0'/2'/2'<0;1>/*))),{and_v(v:after(1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,@3/48'/0'/3'/2'<0;1>/*,@4/48'/0'/0'/2'<0;1>/*,@5/48'/0'/1'/2'<0;1>/*))),{and_v(v:older(65535),multi_a(2,@6/48'/0'/2'/2'<0;1>/*,@7/48'/0'/3'/2'<0;1>/*)),and_v(v:older(4255898),multi_a(1,@8/48'/0'/0'/2'<0;1>/*,@9/48'/0'/1'/2'<0;1>/*,@10/48'/0'/2'/2'<0;1>/*))}}})
```

Two sha256 hashlocks. Both Bitcoin timelock flavours, absolute and relative. `multi_a` at thresholds 3, 2, 2 and 1. A **NUMS internal key**, so there is no key-path spend at all — every route to the coins goes through a script leaf. Eleven cosigners across **three masters**.

Until three commits ago this document could not exist. Every one of its four leaves is `and_v(v:...)` wrapping a timelock or a hashlock, and the device's tap-leaf reader named only `pk / multi_a / sortedmulti_a`. It returned *zero* leaves and the consent screen read "Complex policy — display only", while the host derived the same wallet's addresses without complaint. That was F-214; the fix replaced the vocabulary with an emitter, and this journey is the proof it worked.

The keys are BIP-39 test vectors. Never put funds behind them.

1 · Every slot's origin matches its own key file

The template carries eleven per-key origins, written by hand; the eleven key files declare them independently. If the two disagreed, the card would name a path the key does not live at — and **nothing downstream would notice**, because addresses come from the xpub a card carries, never from the origin it declares. That is F-217's lesson, and this gate is the answer to it.

```
@0 48'/0'/0'/2' ok
@1 48'/0'/1'/2' ok
@2 48'/0'/2'/2' ok
@3 48'/0'/3'/2' ok
@4 48'/0'/0'/2' ok
@5 48'/0'/1'/2' ok
@6 48'/0'/2'/2' ok
@7 48'/0'/3'/2' ok
@8 48'/0'/0'/2' ok
@9 48'/0'/1'/2' ok
@10 48'/0'/2'/2' ok
```

```
0 slot(s) disagree with their key file
```

2 · The engraved artifact

The keyless template set — 4 chunks. No --path: the origins ride in the template, and --path would flatten eleven distinct origins onto one, which over eleven different keys is a card md encode now refuses outright.

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md encode tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,{and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,@0/48'/0'/0'/2'<0;1>/*,@1/48'/0'/1'/2'<0;1>/*,@2/48'/0'/2'/2'<0;1>/*)),{and_v(v:after(1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,@3/48'/0'/3'/2'<0;1>/*,@4/48'/0'/0'/2'<0;1>/*,@5/48'/0'/1'/2'<0;1>/*)),{and_v(v:older(65535),multi_a(2,@6/48'/0'/2'/2'<0;1>/*,@7/48'/0'/3'/2'<0;1>/*)),and_v(v:older(4255898),multi_a(1,@8/48'/0'/0'/2'<0;1>/*,@9/48'/0'/1'/2'<0;1>/*,@10/48'/0'/2'/2'<0;1>/*))}}}) --group-size 0 --force-chunked
chunk-set-id: 0x3d896
md1f8kykps9zjtvyvy5jmplrjttvyvy49ykcgfw2fdsssjttvyvyw2fdssj55jmplp9ef9kzzz2fdsturhn8pf8jq0c
md1f8kykpsgs3ef9kzz2j5qqcr32v6mq85syszv6a4pxuusy2unu8xqz8naa2r2akwukvgm42glxskfar6x4423
md1f8kykpsws7xkx9k7c9xu6pzkjqsspy9fntdcdhkyqfntk5ymnjqjatj0sucqg70h4gdtkec5muezu6y3f47m
md1f8kykpsmje3rw4fp6rc6cckmmq5mngy26gpzx3g4xdwqq8lluszze6v6uqpq0px3qq2y6qqmxcdfn702jwt
note: stdout is a keyless descriptor template (no keys)
[exit 0]
```

```
The engraved set is 4 md1 chunk(s).
```


keyless template-id: 44ad26a19b53048b6ff8957359a30c31
keyed template-id: 44ad26a19b53048b6ff8957359a30c31
SAME WALLET.

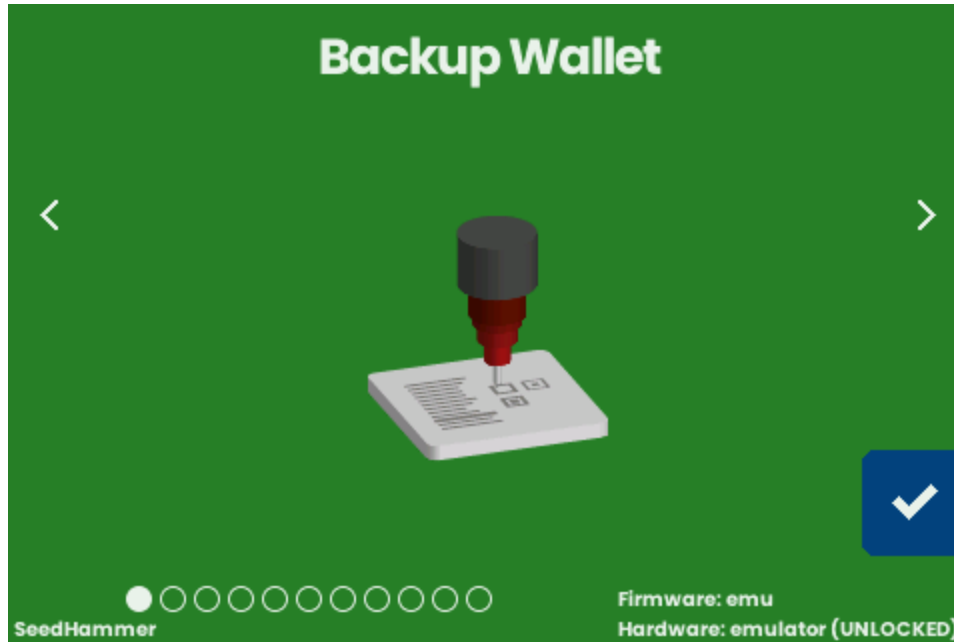
... clipped; full text in transcript_tr_pathological.txt

5 · What the device must prove to

Derived from the *cards*, not from the argument list that made them.

```
wallet idwallet-policy-id: 590f3abcaad2aca5a3f526917f5bb57a  
receive 0..1bc1pgavpejq7pkax6zdtuay9ejenangza2t3l6gfka79h94p23xcexssak7pzz  
bc1p3wy7tcypvkm49wtyeq2yw4t0266vxdnq1kny5nsgzguc0fcvd13s5hkkp5  
change 0..1bc1pnjrzcky5qq1hyz4rvhq5r96ca77wyccnfls0p517k8ae7402vjhshf3zya  
bc1p7yrpv9q6j58vvh5yrnnj8ky6dnykn7z2lak4j0ckt52j84j8fxzs4h6khc
```

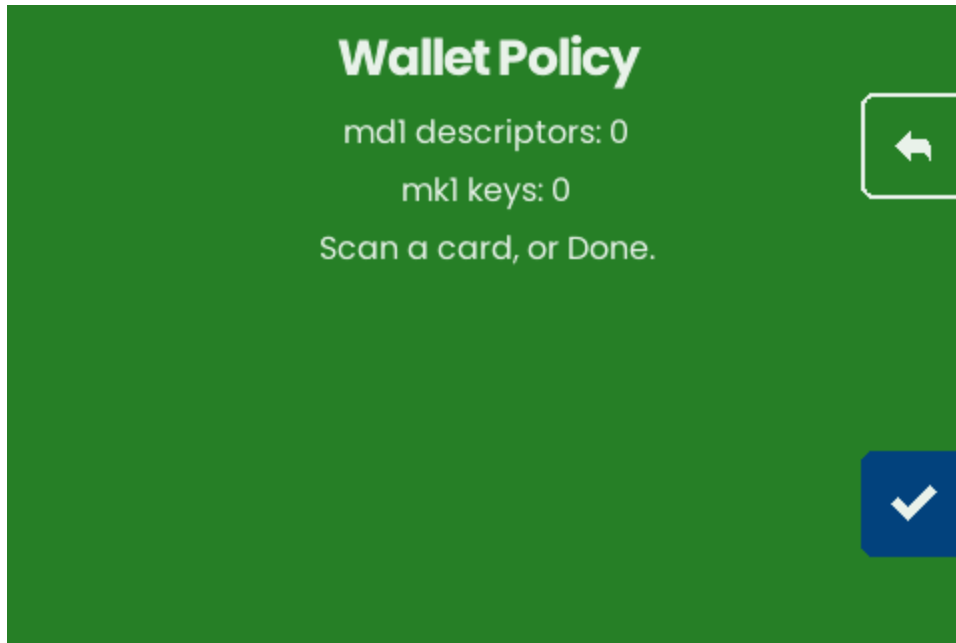
6 · The device



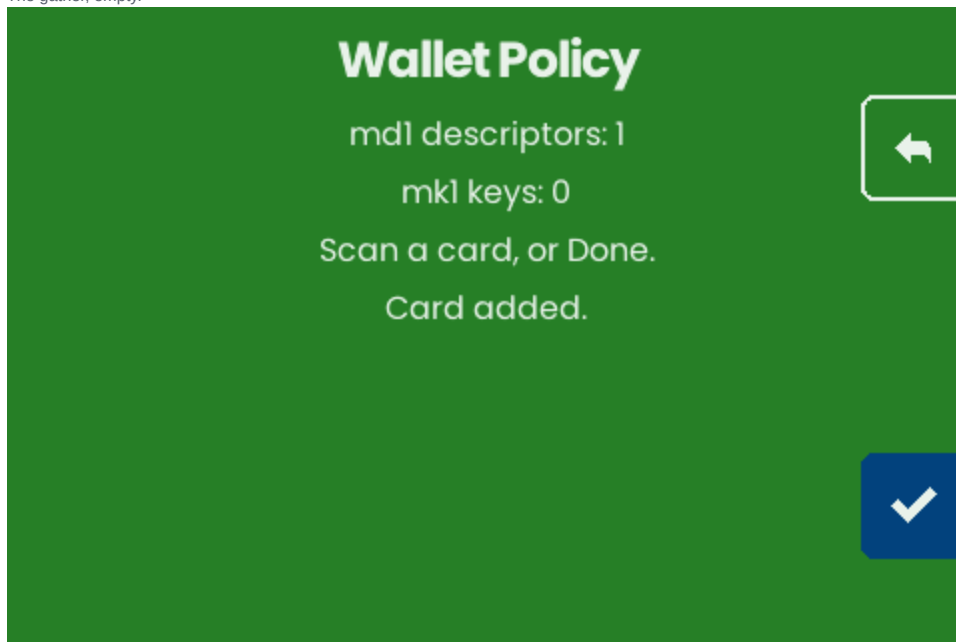
Home.



Wallet Policy.



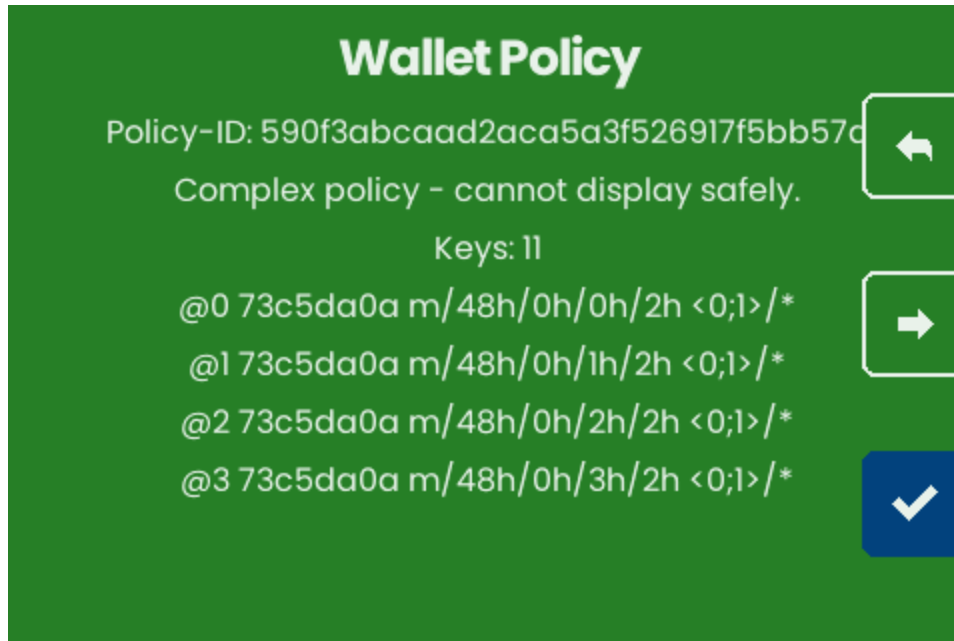
The gather, empty.



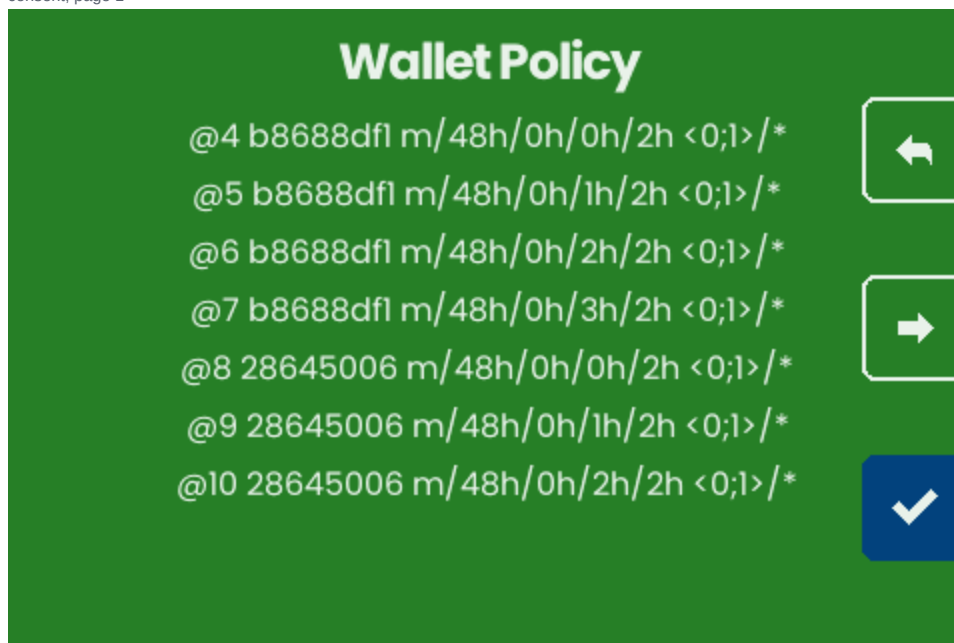
After all 24 chunks cross the reader: ONE card.

7 · The proof, on the consent path

Eleven key slots print before the addresses do, so the screen pages 4 times — and the walk reads every page, because a bound tuned for a four-key wallet would stop before reaching the half that matters.



consent, page 1



consent, page 2

Wallet Policy

Receive 0:
bc1pgavpejq7pkax6zdtuay9ejenangza2t3l6gfka79h94p23xcexssak7pzt

Receive 1:
bc1p3wy7tcypvkm49wtieq2yw4t0266vxdnqlkny5nsgzguc0fcvdl3s5hkkp5

Change 0:

←

→

✓

consent, page 3

Wallet Policy

bc1pnjrzc ky5qqlhyz4rvhq5r96ca77wyccnfls0p5l7k8advjhshf3zya

Change 1:
bc1p7yrpv9q6j58vwh5yrnnj8ky6dnykn7z2lak4j0ckt52xzs4h6khc

←

→

✓

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8 · What this run established

The capture is a **comparison**. It is handed the host's id and addresses and throws if the screen disagrees, and the absence assertions reject "display only" and "no addresses" outright — so the device cannot pass by refusing politely.

```
chunks presented24
cards gathered1
consent pages read4
wallet id matched590f3abcaad2aca5a3f526917f5bb57a
addresses matchedbc1pgavpejq7pkax6zdtuay9ejenangza2t3l6gfka79h94p23xcexsak7pzt
bc1p3wy7tcypvkm49wtieq2yw4t0266vxdnqlkny5nsgzguc0fcvd13s5hkkp5
bc1pnjrzc5qqlhyz4rvhq5r96ca77wyccnfls0p5l7k8ae7402vjhshf3zya
bc1p7yrpv9q6j58vwh5yrnnj8ky6dnykn7z2lak4j0ckt52j84j8fxzs4h6khc
```

The negative control is a command, not a memory. `python3 capture_tr_pathological.py --prove-it-can-fail` corrupts one character of one expected address and exits 0 only if the walk *caught* it. A comparison nobody has made fail is not evidence that it can.

Reproducing

```
bash transcript_tr_pathological.sh > transcript_tr_pathological.txt 2>&1
python3 capture_tr_pathological.py           # rebuilds emu.wasm, drives it
python3 capture_tr_pathological.py --prove-it-can-fail # the negative control
python3 build_pdf_tr_pathological.py         # this document
```

What this journey does NOT show

- **No spend, and for tiers 1–2 no spend is possible from this backup at all.** Both hashlock tiers need the 32-byte preimage to spend, and zero backup strings carry it (F-132).
- **The tiers do not degrade in the order they read.** Tier 4 (1-of-3) matures at 365 days; tier 3 (2-of-2) at ~455 — the weakest key-set unlocks ~90 days *first* (F-133).
- **No plate read-back.** There is no reader; the card-set equality above is the honest proxy.
- **No ms1 secret leg.** Taproot-independent, and already shown in the earlier journeys.
- **The origins are not visible on read-back** (F-219) — the card carries them, and neither `decode` nor `inspect` will show them to an operator.